



NATIONAL OPEN UNIVERSITY OF NIGERIA
PLOT 91, CADASTRAL ZONE, NNAMDI AZIKIWE EXPRESSWAY, JABI-ABUJA
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE

2025_1 EXAMINATION

Course Code: CIT 310

Course Title: Algorithms and Complexity Analysis

Credit: 3 Units

Time allowed: 2½ Hours

Instruction: Answer Questions ONE (1) and any other THREE (3)

Question One

- a. Given the following recursive algorithm:

$$F_0 = 1, F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2}, \text{ for } n \geq 2$$

Find F_7 by simulating it manually **(6Marks)**

- b. Write two (2) pseudo-codes to differentiate between Linear Recursion and Tree recursion **(8Marks)**

- c. Define a function $F(n)$ to calculate the n th Fibonacci number **(4Marks)**

- d. Given that the greatest common divisor, gcd is expressed as:

$$\text{gcd}(p, q) = \begin{cases} p & \text{if } q = 0 \\ \text{gcd}\left(q, \text{remainder}\left(\frac{p}{q}\right)\right) & \text{if } p \geq q \text{ and } q \neq 0 \end{cases}$$

Compute $\text{gcd}(48, 12)$ **(4Marks)**

- e. List three (3) applications of Tower of Hanoi **(3Marks)**

Questions Two

- a. With the help of a diagram, explain indirect recursion **(6Marks)**
b. Explain the principle of Selection Sort **(5Marks)**
c. Briefly describe in steps how Dynamic Programming works **(4Marks)**

Questions Three

- a. Given the Binary Search Tree below with key $[x] = 37$. State the value of key $[y]$ **(4Marks)**
b. Explain the steps to perform Binary Search for '13' from the array of numbers **(6Marks)**
c. State the importance of writing algorithms **(5Marks)**

Question Four

- a. Write a recursive function to calculate n th Fibonacci number **(5Marks)**
b. Write short notes on Randomized Algorithm and mention two types **(8marks)**
c. Explain why the Worst-Case analysis is important in analysis of algorithms **(2Marks)**

Question Five

- a. Using the substitution method, show that $T(n) = T\left(\frac{n}{2}\right) + n$ is asymptotically bounded by $O(\log n)$ **(5Marks)**
b. With a suitable diagram, describe the Omega notation **(6Marks)**
c. Explain two factors used to analyse the efficiency of an algorithm **(4marks)**

Question Six

- a. Given a function $f(n) = n^2 + 7n + 5$.
Explain what it means to say that " $f(n)$ " is **asymptotically equivalent** to n^2 as $n \rightarrow \infty$ " **(6Marks)**
b. Briefly differentiate between an algorithm and a pseudo-code **(6Marks)**
c. Mention and describe the two types of algorithms that are equivalent to each other **(3Marks)**



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Course Code: CIT310
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Credit: 3 units
Time allowed: 3 Hours
Instruction: Answer Questions **ONE (1)** and any other **THREE (3)** Questions

Question One

- a. Identify the major challenges of using a pseudocode in the development computer software. (3Marks)
- b. Describe the major parameters used for the performance analysis of computer algorithms. (4Marks)
- c. What are the main features of a Computer Algorithms. (7 Marks)
- d. Write a binary search pseudocode that returns the position value of number “eleven” in the table below. (5 marks)

Position in data set	0	1	2	3	4	5	6	7	8
Data value	1	3	4	5	7	9	11	14	16

- e. Briefly describe the Complexity of a Binary Search. (6 marks)

Question Two

- a. As a computing student learning the skill of programming, briefly differentiate between an algorithm and a pseudocode. (6Marks)
- b. Recursion is said to be a powerful tool, but it can be a tricky concept to implement. Use python programming language to create a factorial algorithm. (3 marks)
- c. What is the importance of Asymptotic Notation? (2marks)
- d. Write the major advantages and disadvantages associated to the adoption of binary search as an algorithm for solving computational problems. (4 marks)

Question 3

- a. Describe the difference between a recursion and iteration? (6 marks)
- b. Highlight the major difference between a pseudocode and an algorithm. (3 marks)
- c. Use a flow chart from the top, reaching the base case, and then working your way back up to describe the evaluation of 5 factorial (i.e. 5!). (6 marks)

Question Four

- a. Discuss the terms "Sorting in Ascending order" and "sorting in Descending order". (4Marks)
- b. Explain the types of time complexity in an algorithm analysis? (6 marks)
- c. Identify the need of an Algorithms in the field of computer science? (5 marks)

Question Five

- a. Why is stable sort useful? (2 marks)
- b. With examples, illustrate the following Asymptotic Notations.
 - i. Big-oh notation (5 marks)
 - ii. Big Omega (Ω) (4marks)
 - iii. Big Theta (Θ) (4marks)

Question Six

- a. Discuss Prim's Algorithm. (5 marks)
- b. Outline the Steps required for finding Minimum Spanning Tree using Prim's Algorithm (6 marks)
- c. What are the general properties of minimum spanning trees.(4 marks)



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FACULTY OF SCIENCE
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2022_2 EXAMINATION

Course Code: CIT310
Course Title: Algorithms and Complexity Analysis
Credit: 3 units
Time allowed: 3 Hours
Instruction: Answer Questions **ONE (1)** and any other **THREE (3)** Questions

Questions

Question 1

- a. Explain the three laws all recursive algorithms should obey. (3 marks)
- b. Highlight the major difference between a pseudocode and an algorithm. (3 marks)
- c. Discuss the term 'Algorithm'. **(5Marks)**
- d. Explain briefly, how a selection sort works. (7Marks)**
- e. Recursion is said to be a powerful tool, but it can be a tricky concept to implement. Use python programming language to create a factorial algorithm. (3 marks)
- f. Discuss the terms "Sorting in Ascending order" and "sorting in Descending order". **(4Marks)**

Question 2

- a. In a short note briefly discuss the time complexity evaluation of the algorithm below; (8 marks)

```
AO
{
i = 1; s = 1;
while (s<=n)
{
i++;
ss = s + i;
printf("Abdullahi");
}
}
```

- b. Describe the term linear search. **(5Marks)**
c. What is recursion base case? (2 marks)

Question 3

- a. In radix sort algorithm, when does worst case complexity happens? **(3marks)**
b. With examples, illustrate the following Asymptotic Notations.
i. **Big-oh notation (5 marks)**
ii. **Big Omega (Ω) (4marks)**
iii. **Big Theta (θ) (4marks)**

Question 4

- a. Define Red-Black tree, and enumerate its properties. **(5Marks)**
b. Consider the following recursive programs, and discuss the time complexity evaluation of the algorithm. (8 marks)

```
A(n)
{
if (n>1)
return (A(n-1))
}
```

- c. Why is the Worst-case analysis the most important in algorithm analysis? **(2Marks)**

Question 5

- a. There are two types of algorithms that are equivalent to each other, mention and briefly discuss. **(3Marks)**
- b. Highlight ten (10) basic reasons, why algorithms are needed? **(10Marks)**
- c. What is the importance of Asymptotic Notation? (2marks)

Question 6

- a. Describe the difference between a recursion and iteration? (6 marks)
- b. Design a recursive algorithm for reversing the n elements of an array, A, so that the first element becomes the last, the second element becomes the second to the last, and so on. (5 marks)
- c. Consider the Recurrence

$$T(n) = 2T\left(\frac{n}{2}\right) + n \quad n > 1$$

Find an Asymptotic bound on T. (4 marks)



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2024_2 EXAMINATION

Course Code: CIT310
Course Title: Algorithms and Complexity Analysis
Credit: 3 units
Time allowed: 2½ Hours
Instruction: Answer Questions **ONE (1)** and any other **THREE (3)** Questions

- 1a. Enumerate six (6) characteristics of Algorithm **(6Marks)**
- b. Write short notes on Fibonacci sequence **(3Marks)**
- c. Hence, define a function $F(n)$ to calculate the n th Fibonacci number **(4Marks)**
- d. State the advantages and disadvantages of Pseudocode **(6Marks)**
- e. Briefly explain the concept of Linear Search **(6Marks)**
- 2a. With the help of a diagram, explain indirect recursion **(6Marks)**
- b. Given that the greatest common divisor, gcd is expressed as:
$$\gcd(p, q) = \begin{cases} p & \text{if } q = 0 \\ \gcd\left(q, \text{remainder}\left(\frac{p}{q}\right)\right) & \text{if } p \geq q \text{ and } q \neq 0 \end{cases}$$
Compute $\gcd(48, 12)$ **(4Marks)**
- c. Describe the principle of the Tower of Hanoi **(5Marks)**
- 3a. Given three pegs A, B, C, show with illustrative diagrams the 7 steps for transferring all the disks from peg A to peg C in the Tower of Hanoi algorithmic problem **(7Marks)**
- b. Generate a relation formula for total time taken to move n disks from peg A to peg C in the Tower of Hanoi Algorithm **(5Marks)**
- c. List three (3) applications of Tower of Hanoi **(3Marks)**
- 4a. Briefly describe in steps how Dynamic Programming works **(5Marks)**
- b. Write short notes on recurrence relation **(6Marks)**
- c. Explain how bubble sort works **(4Marks)**
- 5a. Write an algorithm to implement Bubble Sort **(7Marks)**

- b. Sort this array of numbers 12 45 23 48 10 in ascending order, using Bubble Sort algorithm **(8Marks)**
- 6a. Enumerate three (3) advantages and disadvantages of Bubble Sort **(6Marks)**
- b. Explain the principle of Selection Sort **(5Marks)**
- c. Write an Algorithm to implement Selection Sort **(4Marks)**



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2022_2 EXAMINATION

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Instruction: Answer Questions **ONE (1)** and any other **THREE (3)** Questions

Questions

Question 1

- a. Explain the three laws all recursive algorithms should obey. (3 marks)
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- f. Discuss the terms "Sorting in Ascending order" and "sorting in Descending order". **(4Marks)**

Question 2

- a. In a short note briefly discuss the time complexity evaluation of the algorithm below; (8 marks)

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A()
{
i = 1; s = 1;
while (s<=n)
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- b. Describe the term linear search. **(5Marks)**
c. What is recursion base case? (2 marks)

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b. With examples, illustrate the following Asymptotic Notations.
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- b. Design a recursive algorithm for reversing the n elements of an array, A, so that the first element becomes the last, the second element becomes the second to the last, and so on. (5 marks)
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- Briefly describe the Complexity of a Binary Search. (6 marks)

Question Two

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2024_2 EXAMINATION

Course Code: CIT310
Course Title: Algorithms and Complexity Analysis
Credit: 3 units
Time allowed: 2½ Hours
Instruction: Answer Questions **ONE (1)** and any other **THREE (3)** Questions

- 1a. Enumerate six (6) characteristics of Algorithm **(6Marks)**
- b. Write short notes on Fibonacci sequence **(3Marks)**
- c. Hence, define a function $F(n)$ to calculate the n th Fibonacci number **(4Marks)**
- d. State the advantages and disadvantages of Pseudocode **(6Marks)**
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- 5a. Write an algorithm to implement Bubble Sort **(7Marks)**

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